

Lab Hour

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Background

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作为校园公益组织负责人，你近期组织了一场二手义卖活动，筹集的全部善款将用于下学期的公益项目。为了深入了解不同学生群体的购买偏好，以便为后续活动做更充分的准备，你计划对活动的销售收入进行分析。然而，由于学生会收集的数据来自多个非标准化渠道，导致众筹记录极其混乱、难以直接使用。

【下载day4lab文件夹，打开day4lab/dataset，浏览这些文档，目前的数据有哪些问题？】

Challenges

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- Inconsistent currency text formatting (100 and ¥100)
- Invalid inputs (negative amounts)
- Whitespaces
- Missing customer's demographic information

Stage 1: Data Cleaning

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Goal:

- Remove invalid records from raw_amount

Tasks:

- Regex: 通过 Regex 移除 \$, ¥ 等货币符号
- Outlier Filtering: 应用filter移除负数

Stage 2: Data Integration

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Goal:

- Merge the income records with student demographics.

Tasks:

- Join `df_stage1` with `df_students` using the key `student_full_name`.

Stage 3: Deductive Imputation

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Goal:

- Fill in missing grade information using deterministic logic
 - By analyzing the adjacent context of the student_id prefix, we can accurately deduce the missing values.
 - Prefix Mapping Rules: S2024 Prefix → Freshman; S2023 Prefix → Sophomore; S2022 Prefix → Junior; S2021 Prefix → Senior

Tasks:

- Complete deduce_grade Function

Stage 4: Data Transformation

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Goal:

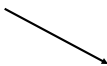
- Scale the raw amounts to a standard range of [0, 1] for proportionate Scaling.

Tasks:

- Find min_val (minimum value) and max_val (maximum value) in the 'amount' column to prepare data for modelling
- Apply the formula $(x - x_{\min}) / (x_{\max} - x_{\min})$ to create a new column, amount_normalized.

What did we do?

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- Inconsistent currency text formatting (100 and ¥100) 
- Invalid inputs (negative amounts) 
- Whitespaces 
- Missing customer's demographic information 

1. Data Cleaning

Handles formatting and outliers.

2. Data Integration

Merges relational tables.

3. Deductive Imputation

Logically infers missing values

4. Data Transformation

Normalizes the metrics.



Cleaned dataset